Optical Simulation of Scintillator Bars
EJ-204 vs EJ-230 for the SHiP Timing Detector

René Ríos   Gerardo Vásquez

SHiP Collaboration — Timing Detector Group

15–16 June 2026

Contents

Bar geometry and SiPM readout

Scintillator bar (GEANT4):

- Active length: 1400 mm (± 700 mm along x)
- Coating: reflective Mylar
- $v_{\text{eff}} \approx 277$ mm/ns (nominal axial average)

↪ ToF onset measures $v_{\text{eff}}^{\text{onset}} \approx 292$ mm/ns ($+5\%$): set by the most direct rays, not the mean path (see QA-1c)

- Two materials compared in parallel: **EJ-204** vs **EJ-230**

End SiPMs (primary timing):

- 16 SiPMs total (IDs 0–15)
- END_LEFT: IDs 0–7 at $x = -700$ mm
- END_RIGHT: IDs 8–15 at $x = +700$ mm
- Timing estimator: **SUM4** ($\min$ over 4-neighbor clusters)

Top SiPMs (veto / tracking):

- 70 SiPMs, single row at $Y = +30.25$ mm
- TOP_LEFT: IDs 16–50 / TOP_RIGHT: 51–85
- Pitch: 20 mm in X; central gap 24 mm

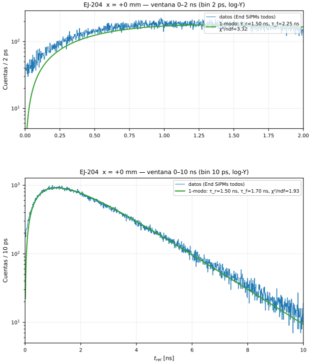
GEANT4 statistics:

Dataset	Events	Positions
EJ-204 End	2000/pos	31
EJ-230 End	2000/pos	31
EJ-204 EndTop	2000/pos	31

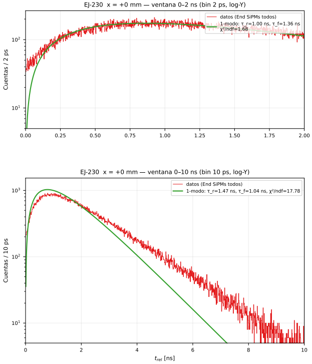
Positions: $x = 0, \pm 50, \ldots, \pm 690$ mm

F1: Relative arrival time t_{rel} — central position $x = 0$

EJ-204 (left) vs EJ-230 (right) — 1-mode fit: $N(1 - e^{-t/\tau_r})\, e^{-t/\tau_f}$



EJ-204 x=+0 mm. $t_{rel} = t_{fotón} - \min(t_{End})$ por evento.



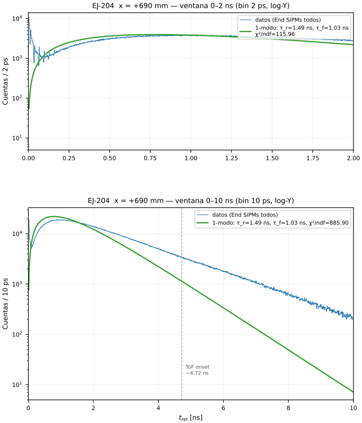
EJ-230 x=+0 mm. $t_{rel} = t_{fotón} - \min(t_{End})$ por evento.

$t_{\text{rel}} = t_{\text{photon}} - \min(t_{\text{End}})$ per event. EJ-204: $\tau_r = 1.50$ ns, $\tau_f = 1.70$ ns ($\chi^2/\text{ndf} = 1.93$). EJ-230: $\tau_r = 1.47$ ns, $\tau_f = 1.04$ ns ($\chi^2/\text{ndf} = 17.8$).

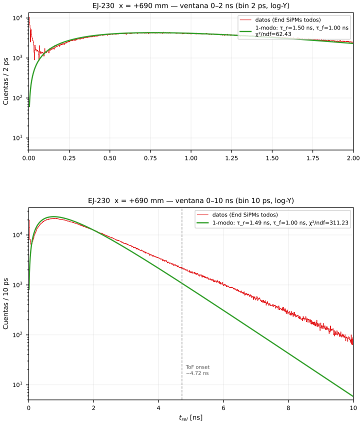
Fitted τ describe the arrival-time profile (emission $\otimes$ propagation), NOT the datasheet emission constants (EJ-204: $\tau_r^{(0)} = 0.7$ ns, $\tau_d^{(0)} = 1.8$ ns).

EJ-230 tail at $x = 0$: **cannot** be ToF (ends equidistant); origin undetermined — slow mode vs reflections.

Second component at ~ 4.7 ns visible in both materials



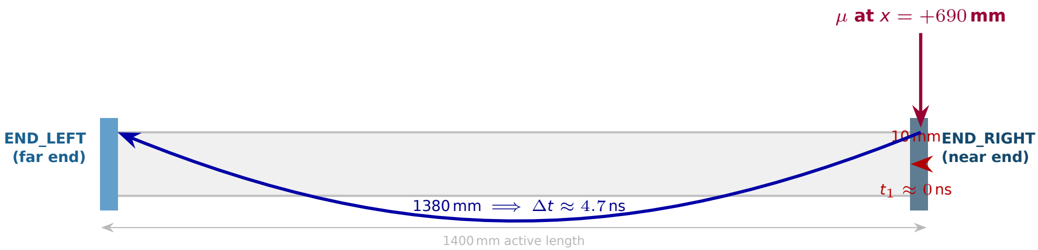
EJ-204 x=+690 mm. $t_{rel} = t_{fotón} - \min(t_{End})$ por evento.
Cola 3-10 ns: ~98% fotones cercano dispersados + ~2% ToF lejano (QA-1c onset~4.72ns).



EJ-230 x=+690 mm. $t_{rel} = t_{fotón} - \min(t_{End})$ por evento.
Cola 3-10 ns: ~98% fotones cercano dispersados + ~2% ToF lejano (QA-1c onset~4.72ns).

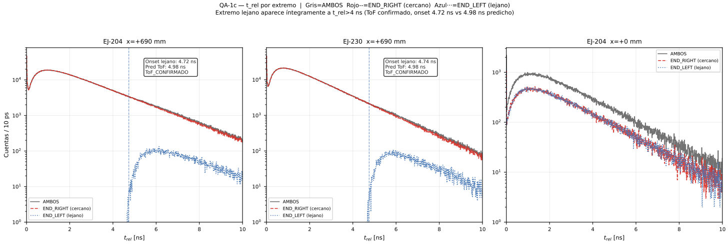
1-mode fit fails at $x = +690$: $\chi^2/\text{ndf} = 886$ (EJ-204), 721 (EJ-230). Tail 3–8 ns: **not** slow scintillation — see ToF diagram below and QA-1c.

Why a second component appears at the edge — geometric time-of-flight



The late bump is the *same* photons reaching the *far end* — not slow scintillation.

QA-1c: Second F1 component confirmed as geometric ToF



ToF prediction ($x = +690$ mm):

$$\Delta t_{\text{pred}} = \frac{1380\,\text{mm}}{277\,\text{mm/ns}} \approx 4.98\,\text{ns}$$

Measured onset (far end):

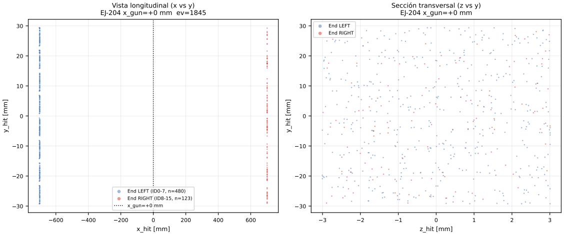
- EJ-204: 4.72 ns (-5.3%)
- EJ-230: 4.74 ns (-4.9%)

Verdict:
ToF_CONFIRMED

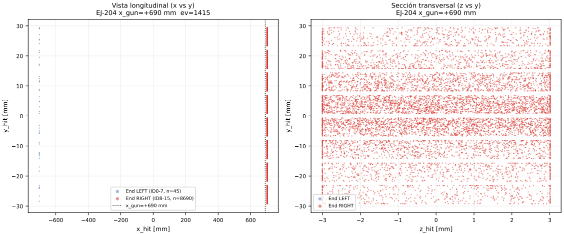
The -5% offset reflects $v_{\text{eff}}^{\text{onset}} \approx 292$ mm/ns — the fastest (most direct) optical rays arrive first, pulling the onset earlier than the mean-path prediction.

F3: Spatial distribution of photon impacts on SiPMs

Single representative event — EJ-204 (EJ-230 qualitatively identical)



F3 — Impactos de fotones en SiPMs (NO trazas en volumen), evento 1845 (603 hits). EJ-204 evento display reconstruido desde hits (NO re-simulacion).



F3 — Impactos de fotones en SiPMs (NO trazas en volumen), evento 1415 (8735 hits). EJ-204 evento display reconstruido desde hits (NO re-simulacion).

$x_{\text{gun}} = 0$ mm — symmetric L/R distribution

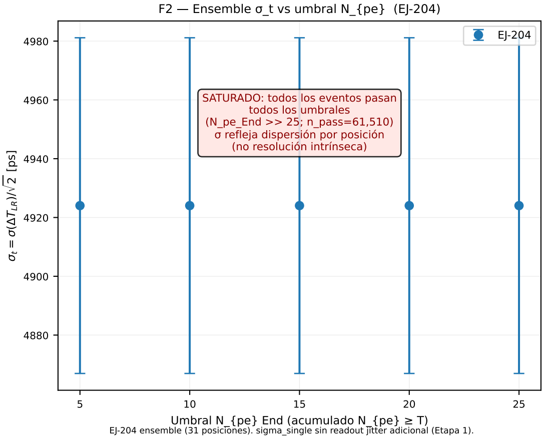
$x_{\text{gun}} = +690$ mm — collapse toward END_RIGHT

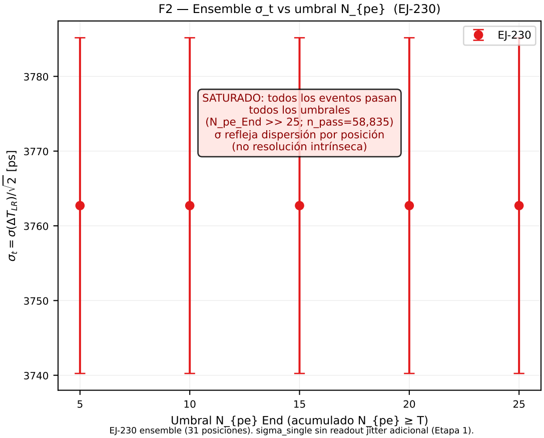
Note: Points are *photon impacts on the SiPM window* (NOT trajectories inside the scintillator volume). Proxy of optical propagation.

Reconstructed from `x_mm/y_mm/z_mm` ROOT branches — no re-simulation.

F2: Ensemble σ_t vs End N_{pe} threshold

Both materials saturated at 100% — threshold *does not* discriminate

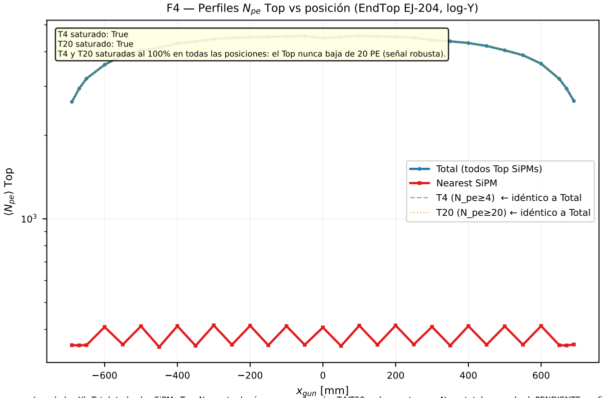




EJ-204 **EJ-230**

All events pass thresholds 5–25 PE ($N_{\mathrm{pe}}^{\mathrm{End}} \gg 25$). $\sigma \approx 5$ ns reflects **positional spread** across 31 positions, not intrinsic resolution. Per-position resolution: see F5.

F4: $\langle N_{pe} \rangle$ Top profiles vs position (EJ-204 EndTop)

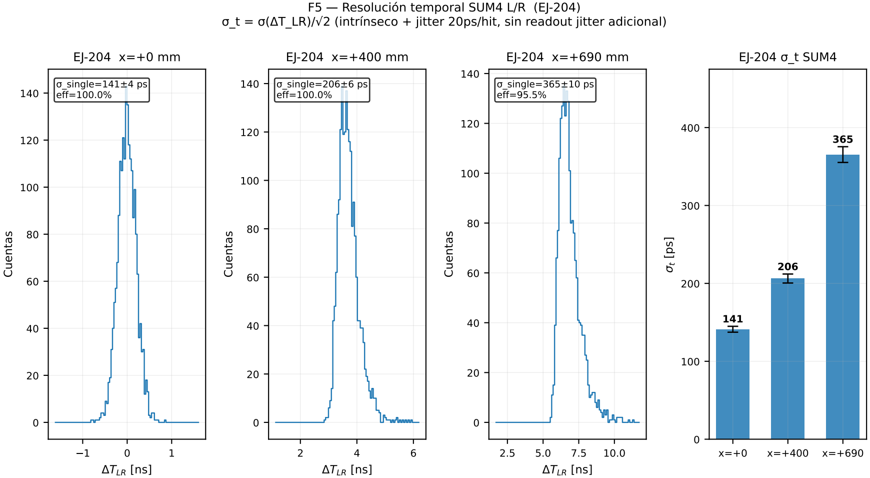


EJ-204 (EndTop) perfiles N_pe Top vs x_gun (escala log-Y). Total: todos los SiPMs Top. Nearest: el más cercano al muón. T4/T20: solo eventos con N_pe_total >= umbral. PENDIENTE confirmación Gerardo sobre definición T4/T20.

- T4 and T20 **saturated at 100%** at all positions: Top never falls below 20 PE — robust signal across the full kinematic range.
- “Nearest” curve oscillates with the discrete 20 mm SiPM pitch.
- **Open — Gerardo:** Is T4/T20 a cut on total N_{pe} (as implemented) or a SUM4 trigger efficiency?

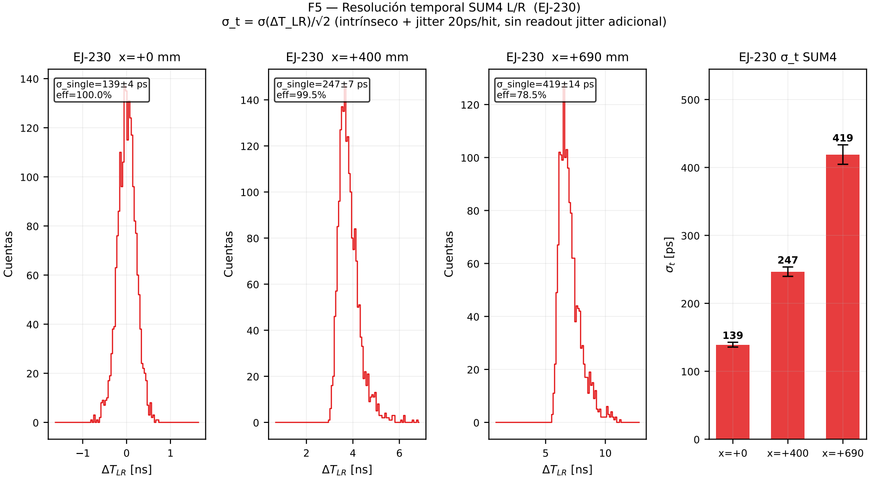
F5: SUM4 L/R timing resolution — EJ-204

$\sigma_t = \sigma(\Delta T_{LR})/\sqrt{2}$; jitter 20 ps/hit (Stage 1); no time-walk or readout jitter



F5: SUM4 L/R timing resolution — EJ-230

$\sigma_t = \sigma(\Delta T_{LR})/\sqrt{2}$; same Stage 1 caveats — stronger positional degradation than EJ-204



Material	$x = 0$ mm σ_t [ps]	$x = +400$ mm σ_t [ps]	$x = +690$ mm σ_t [ps]
EJ-204	141 ± 4	206 ± 6	365 ± 10
EJ-230	139 ± 4	247 ± 7	419 ± 14

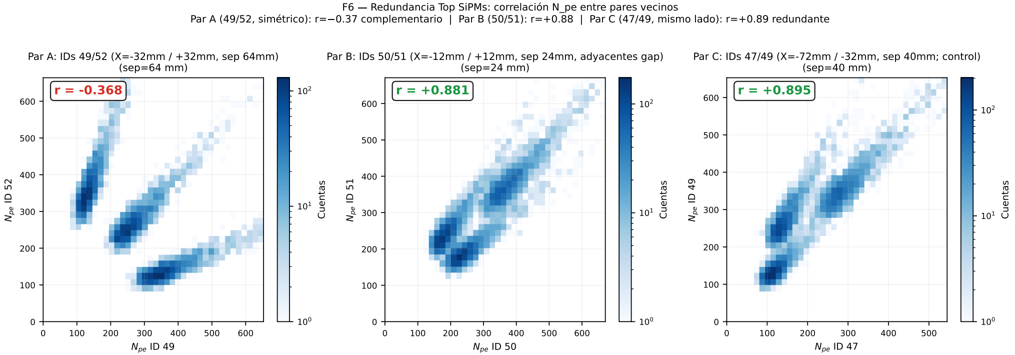
Key observations:

- Central resolution **equivalent**: $141 \approx 139$ ps
- EJ-204 $\sim 15\%$ better at the edge (365 vs 419 ps)
- EJ-230 degrades faster with position ($+201\%$ vs $+159\%$ from $x = 0$ to $x = 690$)
- Fitted τ_f: arrival-time profile (emission $\otimes$ propagation)
  not datasheet constants
  (EJ-204: $\tau_r^{(0)} = 0.7$ ns, $\tau_d^{(0)} = 1.8$ ns)

Caveats — Stage 1 (no real electronics):

- Jitter 20 ps/hit: possible double-count (L+R)
- No time-walk correction (ToT)
- No readout jitter applied
- SUM4 topology: pending Gerardo
- Recovering emission constants requires deconvolution — Stage 2

F6: Top SiPM redundancy — N_{pe} correlation between neighbors

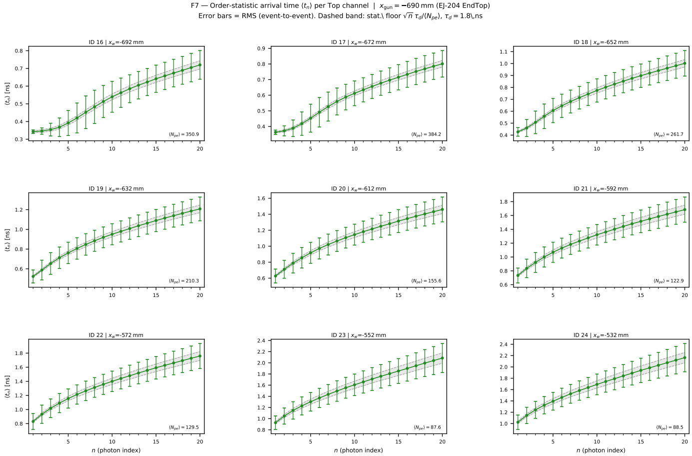


Pair A (IDs 49/52)
sep $= 64\,$mm, opposite sides
$r = -0.37$ — complementary
(light splits between sides)

Pair B (IDs 50/51)
sep $= 24\,$mm, across gap
$r = +0.88$ — **redundant**
(same light source)

Pair C (IDs 47/49)
sep $= 40\,$mm, same side
$r = +0.90$ — **control**
(local correlation expected)

F7: Order-statistic arrival time $\langle t_n \rangle$ per Top channel — $x = -690$ mm
Statistical floor $\sqrt{n}\,\tau_d/\langle N_{\mathbf{pe}}\rangle$ ($\tau_d = 1.8$ ns, EJ-204 datasheet); gap = optical path dispersion



What the axes show:

- x-axis: n = index of the n-th detected photon (order statistic)
- y-axis: $\langle t_n \rangle$ [ns] — mean arrival time per event
- Error bars: RMS over events
- Band: statistical floor $\sigma_{\text{stat}}(t_n) = \sqrt{n}\,\tau_d/\langle N_{\text{pe}}\rangle$

Physics:

RMS $- \sigma_{\text{floor}}$ isolates **optical path dispersion** — the same spread that broadens σ_t and shifts $v_{\text{eff}}^{\text{onset}}$ vs nominal.

At $x = -690$: cluster uses IDs 16–24 ($x_w = -692 \to -532$ mm) — truncated at the left edge; no channels further left.

Summary and open questions

Main results:

1. **EJ-204** $\approx$ **EJ-230** at center (141 vs 139 ps); EJ-204 **better at edges** (365 vs 419 ps at $x = 690$)
2. F1 component at ~ 4.7 ns at $x = 690$ = **geometric ToF** from far end (QA-1c, $\delta = -5\%$). Not slow scintillation.
3. $v_{\text{eff}}^{\text{nominal}} = 277$ mm/ns; onset gives $v_{\text{eff}}^{\text{onset}} \approx 292$ mm/ns ($+5\%$) — fastest optical paths set the onset
4. F7: $\langle t_n \rangle$ RMS exceeds the statistical floor (order-statistics limit) — quantifies optical path dispersion; same physics as σ_t degradation toward edges
5. Top: robust > 20 PE; gap-adjacent SiPMs highly correlated ($r \approx 0.88$)

Open — Gerardo confirmation:

- ☐ SUM4 topology: winner-takes-first vs cluster sum
- ☐ T4/T20: N_{pe} cut or trigger efficiency
- ☐ Jitter 20 ps: single or double-count (L+R)
- ☐ F3: SiPM impacts vs volume trajectories
- ☐ Pair A ($r = -0.37$): geometric or reflections
- ☐ EJ-230 tail at $x = 0$: slow mode or reflections?

Next steps:

- EJ-230 EndTop dataset (Top response)
- Stage 2: time-walk, readout jitter, real electronics
- Deconvolution: emission constants from arrival profiles